

# NUMERICAL ON SENSORS

A capacitive transducer consist of two parallel plate of diameter 2 cm each separated by air gap of 0.25 mm. Assuming a linear response calculate displacement sensitivity of sensor?

Soln:

The displacement sensitivity =  $\frac{dC}{dX}$  , Capacity,  $C = \frac{\epsilon_0 A}{X}$

$$\frac{dC}{dX} = \frac{\epsilon_0 A}{X^2}$$

$$= \frac{8.854 \times 10^{-12} \times 3.14 \times 10^{-4}}{(0.25)^2 \times 10^{-6}} = 444 \times 10^{-10} \text{ F/m}$$

Where:

$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$  (permittivity of air)

A = area of plate

d = separation between plates



# **NUMERICAL ON SENSORS**

**A hydraulic press used in metal forming is monitored using a pressure sensor.**

**The sensor outputs:**

**20 mV at 5 bar**

**80 mV at 20 bar**

**During operation, the measured output is 50 mV.**

- a) Calculate the sensitivity of the pressure sensor.**
- b) Estimate the hydraulic pressure inside the system.**
- c) Comment on whether the press is operating within a 15 bar safety limit.**

# NUMERICAL ON SENSORS

**Solution.**

**Given: 20 mV at 5 bar and 80 mV at 20 bar**

**Therefore,**

$$S = \frac{80 - 20}{20 - 5} = 4 \text{ mv/bar}$$

**The sensitivity of sensor is mv/bar**

## (b) Estimate hydraulic pressure at 50 mV

First find how much the output has increased from the lower point:

$$\Delta V = 50 - 20 = 30 \text{ mV}$$

Corresponding pressure increase:

$$\Delta P = \frac{\Delta V}{S} = \frac{30}{4} = 7.5 \text{ bar}$$

Add to the starting pressure (5 bar):

$$P = 5 + 7.5 = 12.5 \text{ bar}$$

**c) Since 12.5 bar < 15 bar → Safe operation**

**A wearable glucose biosensor is used for continuous blood glucose monitoring.**

**The biosensor produces 25  $\mu\text{A}$  at a glucose concentration of 5 mM.**

**The background noise current is 1  $\mu\text{A}$ .**

- a) Calculate the sensitivity of the glucose sensor.**
- b) Estimate the limit of detection (LOD) of the biosensor.**
- c) Comment on whether the sensor is suitable for detecting low glucose levels ( $\sim 0.7$  mM).**

**Given:**

25  $\mu\text{A}$  at 5 mM and Noise = 1  $\mu\text{A}$

**Solution:**

a) Sensitivity,  $S = \frac{25}{5} = 5 \text{ } \mu\text{A}/\text{mM}$

**b) Limit of Detection:**

$$LOD = \frac{3 \times (\text{noise})}{S}$$

$$LOD = \frac{3 \times 1 \mu\text{A}}{5 \mu\text{A}/\text{mM}}$$

$$LOD = 0.6 \text{ mM}$$

$$\boxed{LOD \approx 0.6 \text{ mM}}$$

**(c) Suitability for detecting ~0.7 mM glucose**

Since:

$$0.7 \text{ mM} > 0.6 \text{ mM}$$

The glucose level of 0.7 mM is above the detection limit.

**An environmental agency uses an amperometric dissolved oxygen (DO) sensor to monitor river health.**

**Calibration data:**

**10  $\mu\text{A}$  at 2 mg/L**

**46  $\mu\text{A}$  at 10 mg/L**

**A field measurement gives a sensor current of 28  $\mu\text{A}$ .**

**a) Determine the DO concentration in the river.**

**b) If healthy aquatic life requires DO > 5 mg/L, comment on the water quality.**

**Given: Calibration data:**

**10  $\mu$ A at 2 mg/L    46  $\mu$ A at 10 mg/L**

$$\text{sensitivity, } S = \frac{46 - 10}{10 - 2} = 4.5 \text{ } \mu\text{A per mg/L}$$

**DO at 28  $\mu$ A:**

$$\text{DO} = \frac{28 - 10}{4.5} + 2 = 6 \text{ mg/L}$$

**A capacitive soil moisture sensor is used in an automated irrigation system.**

**The sensor shows:**

**200 pF at 0% moisture**

**500 pF at 60% moisture**

**During field operation, the measured capacitance is 350 pF.**

**a) Calculate the soil moisture content.**

**b) Should irrigation be activated if optimal soil moisture is below 30%?**

**Given: 200 pF at 0% moisture (baseline) and 500 pF at 60% moisture**

**Solution:**

$$\text{Sensitivity, } S = \frac{500-200}{60-0} = 5 \text{ pF/ \%Moisture}$$

**soil moisture content at measure capacitance of 350 pF**

$$\text{Soil moisture} = \frac{350-200}{5} = 30 \text{ \% Moisture}$$

**A CO<sub>2</sub> sensor is installed in a smart classroom to monitor air quality.**

**Sensor calibration:**

**2.2 V at 400 ppm**

**3.8 V at 1000 ppm**

**During a lecture, the sensor output reads 3.0 V.**

- a) Estimate the CO<sub>2</sub> concentration in the room.**
- b) If recommended indoor CO<sub>2</sub> levels are < 800 ppm, comment on ventilation adequacy.**

**Solution:**

$$\text{Sensitivity, } S = \frac{3.8-2.2}{1000-400} = 0.00267 \text{ V/ppm}$$

**CO<sub>2</sub> concentration when output is**

$$\text{Conc.} = \frac{3-2.2}{0.00267} + 200 = 700 \text{ ppm}$$

**Since the CO<sub>2</sub> levels is < 800 ppm, air quality is acceptable**

**A turbidity sensor monitors water clarity after filtration.**

**Calibration:**

**1.5 V at 10 NTU**

**4.5 V at 80 NTU**

**The sensor shows 3.0 V at the outlet.**

**a) Determine the turbidity of treated water.**

**b) If potable water standards require turbidity  $< 5$  NTU, comment on water quality.**

**Given:**

**1.5 V at 10 NTU    4.5 V at 80 NTU**

**Solution:**

$$\text{Sensitivity, } S = \frac{4.5-1.5}{80-10} = 0.043 \text{ V/NTU}$$

**At 3.0 V:**

$$\text{Turbidity} = \frac{3-1.5}{0.043} + 10 = 45 \text{ NTU}$$

**water quality is not acceptable for potable water.**

**An inductive proximity sensor detects metal parts on a conveyor.**

**Sensor output voltage varies linearly:**

**2 V at 2 mm**

**10 V at 10 mm**

**Measured output = 6 V**

**Questions:**

- a) Determine sensor sensitivity (V/mm)**
- b) Calculate distance of metal object**
- c) If detection range is 8 mm, will the object be detected?**